

BOTANY HUB MULTAI

B.Sc. I BOTANY — MINOR PAPER II

DIVERSITY OF PLANTS

UNIT II — Algae and Fungi

DETAILED TEACHER-STYLE NOTES

Definitions • Complete explanations • Classification tables • Life cycles • Flowcharts • Diagrams • Examples • Economic importance • Exam points

Prepared strictly from the uploaded 2025–26 Minor Paper II syllabus.

UNIT II — ALGAE AND FUNGI

This unit covers algae: characteristics, thallus organization, Fritsch classification, economic importance, and Volvox; fungi: characteristics, classification, economic importance, and Mucor. ■filecite■turn11file0■L48-L56■

A. ALGAE

Definition — Algae: A diverse assemblage of predominantly aquatic, photosynthetic organisms with relatively simple thalloid bodies; they are studied traditionally in phycology.

General characteristics

- Mostly aquatic or found in moist habitats.
- Body commonly forms a thallus without true roots, stems and leaves.
- Photosynthetic pigments vary among groups.
- Reproduction may be vegetative, asexual or sexual.
- Reserve food and cell-wall composition vary among groups.

Thallus organization

Organization	Meaning	Example idea
Unicellular	Single independent cell	Chlamydomonas-type
Colonial	Many cells associated in a colony	Volvox
Filamentous	Cells arranged in threads	Spirogyra-type
Siphonaceous/coenocytic	Large multinucleate body without complete cross walls	Vaucheria-type
Parenchymatous	Tissue-like organization	Large seaweeds

Fritsch classification — exam-level outline

Group	Common textbook basis	Representative examples
Chlorophyceae	Green algae; chlorophyll a and b	Chlamydomonas, Volvox, Spirogyra
Xanthophyceae	Yellow-green algae	Vaucheria
Chrysophyceae	Golden-brown forms	Selected chrysophytes
Bacillariophyceae	Diatoms with siliceous walls	Navicula
Cryptophyceae	Cryptomonads	Cryptomonas
Dinophyceae	Dinoflagellates	Ceratium
Phaeophyceae	Brown algae	Ectocarpus, Laminaria
Rhodophyceae	Red algae	Polysiphonia
Euglenophyceae	Euglenoid forms	Euglena

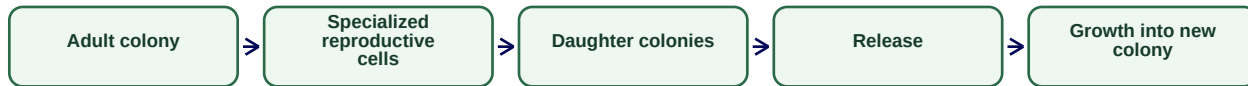
Economic importance of algae

- Food and feed resources.
- Hydrocolloids such as agar, alginates and carrageenan from suitable groups.
- Diatomaceous earth from fossil diatom deposits has industrial uses.
- Primary production and oxygen generation in aquatic ecosystems.
- Some algae form harmful blooms under favorable conditions.

Volvox — structure and reproduction

Definition — Volvox: A colonial green alga in which many flagellated cells are arranged in a hollow spherical colony embedded in a gelatinous matrix.

Asexual reproduction in Volvox



Volvox may reproduce asexually through formation of daughter colonies and sexually through fusion of gametes. The exact sexual condition differs among species. The life cycle is commonly explained with a dominant haploid vegetative phase and a diploid zygote that undergoes meiosis during germination, restoring haploid cells.

B. FUNGI

Definition — Fungi: Eukaryotic, non-photosynthetic organisms that obtain nutrients by absorption and generally possess chitin-containing cell walls; the body is often composed of hyphae forming a mycelium.

General characteristics

- No chlorophyll; nutrition is absorptive.
- Hyphae may be septate or aseptate/coenocytic depending on group.
- Reserve materials commonly include glycogen and lipids.
- Reproduction may be vegetative, asexual and sexual.
- Fungi can be saprophytic, parasitic or mutualistic.

Classification — broad teaching outline

Traditional broad group	Key feature	Example
Phycomycete-type forms	Coenocytic hyphae in many textbook examples	Mucor
Ascomycete-type	Asci associated with sexual spores	Yeast, Penicillium
Basidiomycete-type	Basidia associated with sexual spores	Mushrooms
Deuteromycete/mitosporic concept	Asexual stage emphasized in older systems	Several imperfect-fungus examples

Economic importance

- Decomposers recycle nutrients.
- Fermentation and food industries use selected fungi.
- Some produce useful metabolites and medicines.
- Some cause plant, animal and human diseases.
- Some spoil food or produce harmful toxins.

Mucor — structure, reproduction and life cycle

Definition — Mucor: A fast-growing saprophytic mould with a coenocytic mycelium; asexual reproduction commonly occurs by sporangiospores formed inside sporangia.

Asexual cycle of Mucor



Sexual reproduction in compatible strains leads to fusion of gametangial structures and formation of a resistant zygosporangium. On germination, meiosis restores the haploid phase and a germ sporangium can produce spores.

Exam focus: High-yield: Algae definition + thallus forms + Fritsch table + Volvox life cycle; Fungi definition + classification + economic importance + Mucor asexual and sexual reproduction. Use separate headings and a flowchart for every life cycle.